Bijective Counting, Stars & Bars, PIE & Pigeonhole Practice Exercises

1 How many ordered 3-tuples of integers $(x_1, \ldots, x_3)$ satisfy $x_1 + \ldots + x_3 = 9$ subject to $x_i \geq 1$ for every $i = 1, \ldots, 3$?

a. 55

b. 84

c. 28

d. 25

2 How many ordered 4-tuples of integers $(x_1, \ldots, x_4)$ satisfy $x_1 + \ldots + x_4 = 15$ subject to $x_i \geq 2$ for every $i = 1, \ldots, 4$?

a. 816

b. 120

c. 330

d. 116

3 How many ordered 5-tuples of integers $(x_1, \ldots, x_5)$ satisfy $x_1 + \ldots + x_5 = 13$ subject to $x_i \geq 1$ for every $i = 1, \ldots, 5$?

a. 495

b. 2380

c. 1287

d. 490

4 How many ordered 3-tuples of integers $(x_1, \ldots, x_3)$ satisfy $x_1 + \ldots + x_3 = 15$ subject to $x_i \geq 2$ for every $i = 1, \ldots, 3$?

a. 136

b. 220

c. 52

d. 55

5 How many integers in $\{1, 2, \ldots, 180\}$ are divisible by at least one of 2, 3, or 5?

a. 48

b. 138

c. 132

d. 128

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6 How many integers in $\{1, 2, \ldots, 210\}$ are divisible by at least one of 2, 3, or 7?

a. 60

b. 150

c. 155

d. 146

7 How many integers in $\{1, 2, \ldots, 280\}$ are divisible by at least one of 2, 5, or 7?

a. 184

b. 96

c. 188

d. 180

8 How many integers in $\{1, 2, \ldots, 315\}$ are divisible by at least one of 3, 5, or 7?

a. 144

b. 174

c. 167

d. 171

9 Using Burnside's Orbit-Counting Lemma, how many rotationally distinct necklaces of 6 beads can be formed using up to 2 colors (where two necklaces are identical if one is a rotation of the other, without flipping)?

a. 11

b. 16

c. 14

d. 12

10 Using Burnside's Orbit-Counting Lemma, how many rotationally distinct necklaces of 6 beads can be formed using up to 3 colors (where two necklaces are identical if one is a rotation of the other, without flipping)?

a. 122

b. 130

c. 133

d. 127

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11 Using Burnside's Orbit-Counting Lemma, how many rotationally distinct necklaces of 8 beads can be formed using up to 2 colors (where two necklaces are identical if one is a rotation of the other, without flipping)?

a. 36

b. 32

c. 38

d. 34

12 Using Burnside's Orbit-Counting Lemma, how many rotationally distinct necklaces of 9 beads can be formed using up to 2 colors (where two necklaces are identical if one is a rotation of the other, without flipping)?

a. 57

b. 62

c. 58

d. 60

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